Gradient-Boosted Survival Analysis for Predicting Time to Osteoradionecrosis After Head and Neck Radiotherapy

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Section 1: Describing the Problem and the Dataset

Context. Osteoradionecrosis (ORN) is a late complication of head and neck radiotherapy (RT) in which irradiated bone, most often the mandible due to its dense cortical structure and tenuous blood supply, fails to heal, leading to exposed necrotic bone, pain, infection, pathologic fracture, and substantial loss of quality of life [1, 2]. Reported rates vary widely (from under 5% to over 20%) because of differences in diagnostic criteria, RT technique, and the pooling of anatomic subsites [3]. A defining feature of ORN is its variable latency: it may appear months or many years after treatment, and many patients never develop it at all. This makes the clinically useful question not simply *whether* a patient will develop ORN, but *how long until ORN, if ever*, which is a time-to-event problem. Accurate risk stratification at the time of RT could guide the intensity of dental surveillance, mandibular dose constraints, and patient counseling.

Research question. Can baseline clinical and treatment characteristics, available at the time of RT, predict the time to ORN in a head and neck cancer (HNC) cohort, and does a flexible tree-based survival model improve discrimination over a standard linear Cox model?

Dataset. We used de-identified electronic health record data from the UCSF Clinical Data Warehouse. The analysis cohort consisted of HNC patients who received curative- or palliative-intent RT and whose first RT course was captured in the institutional legacy oncology system (the "AllMQ" source, spanning 2012 onward), which provides the longest and most complete follow-up. ORN events were ascertained by a chart-validated identification methodology developed for a parallel project, combining encounter-level diagnosis text and ICD-coded osteoradionecrosis diagnoses. After excluding records lacking a survival anchor or baseline demographics, the cohort comprised **1,347 patients, of whom 74 (5.5%) developed ORN**; the remaining 94.5% were right-censored. Median follow-up was 51 months and the median index RT dose was 66 Gy.

Each patient contributes a duration and an event status. The duration is the time from RT start to the end of observation: time-to-ORN for cases, and time to the last recorded encounter across *all* hospital departments (or death) for non-cases. Candidate predictors, all fixed at or before the index RT course, included age, sex, primary tumor site (grouped into anatomic subsites from mixed ICD-9 and ICD-10 codes), index RT dose, number of fractions, RT duration, cumulative dose across courses, number of RT courses, re-irradiation status, and RT technique (IMRT/VMAT vs. other). Variables derived from the outcome were excluded to prevent leakage. Patient characteristics stratified by ORN status are presented in Table 1; observed ORN rates were highest in salivary, nasopharyngeal, and oral cavity primaries and near zero in laryngeal primaries, consistent with the degree to which the mandible lies within the high-dose region.

Table 1. Characteristics of the AllMQ HNC RT cohort (n = 1,347), stratified by osteoradionecrosis (ORN) status. Continuous variables are medians; categorical variables are percentages. Follow-up is from RT start to ORN, death, or last all-department encounter.

Characteristic	No ORN (n = 1,273)	ORN (n = 74)
Age at RT, median (y)	63.2	59.8
Female, %	29.4	33.8
Index RT dose, median (Gy)	66.0	66.0
Fractions, median	30	32
Re-irradiation, %	9.2	9.5
Follow-up, median (mo)	51.1	40.6

Observed ORN rate by primary site: salivary 10.5%, nasopharynx 10.3%, oral cavity 7.7%, nasal/sinus 5.0%, bone 4.7%, oropharynx 4.2%, hypopharynx 2.5%, nodal/unknown 2.3%, larynx 0.0%.

Section 2: Describing Gradient-Boosted Survival Analysis

Survival analysis and censoring. Survival analysis models the time T until an event. It is described by the survival function $S(t) = P(T > t)$, the probability of remaining event-free past time t, and the hazard $h(t)$, the instantaneous event rate at t among those still event-free. The central difficulty is *right-censoring*: at analysis time most patients have not had the event, and we know only that they were event-free up to their last contact. Labeling a censored patient simply as "no ORN" is incorrect (they may fail later) and discards the time information. Survival methods are built to learn from these partial observations, using an event indicator $\delta_i \in \{0, 1\}$ alongside the observed time t_i for each patient i.

The Cox model. The Cox proportional hazards model [4] is the standard backbone. It writes each patient's hazard as a shared, unspecified baseline $h_0(t)$ scaled by an exponentiated risk score that depends on the covariates x:

$$h(t \mid x) = h_0(t)\ \exp\big(f(x)\big).$$

In the classical model the risk score $f(x) = \beta^\top x$ is linear. The parameters are estimated by maximizing the *partial likelihood*, which depends only on the order in which events occur and therefore sidesteps the unknown baseline $h_0(t)$. Writing $R(t_i)$ for the risk set (patients still event-free and uncensored just before t_i), the partial likelihood loss to be minimized is the negative log partial likelihood:

$$\mathcal{L} = -\sum_i \delta_i \left[f(x_i) - \log \sum_{j \in R(t_i)} \exp\big(f(x_j)\big) \right].$$

Boosting the risk score. Gradient boosting [5] builds the risk score as an additive ensemble of shallow regression trees rather than a single linear formula. Starting from an initial estimate, it repeats a simple loop for $t = 1, 2, \ldots, M$:

1. Compute the negative gradient of the loss with respect to the current prediction at each patient, $g_i = -\partial\mathcal{L}/\partial f(x_i)$;
2. Fit a regression tree b_t to these pseudo-residuals g_i;
3. Update the model with a learning-rate-scaled correction,

$$f^{(t)}(x) = f^{(t-1)}(x) + \lambda\, b_t(x), \qquad 0 < \lambda \leq 1.$$

The only piece that changes relative to the gradient-boosted classifier covered in lecture is the loss. For binary classification the loss is cross-entropy and the negative gradient reduces to the residual $y_i - p_i$ (label minus predicted probability). For survival, we substitute the negative log partial likelihood above; its negative gradient has a clean interpretation as a *martingale residual*,

$$g_i \;=\; \delta_i - \widehat{H}_i \;\approx\; (\text{observed events}) - (\text{expected events}),$$

where $\widehat{H}_i$ is the patient's estimated cumulative hazard. Each tree is thus fit to the discrepancy between how many events a patient was observed to have and how many the current model expected, which follows the boosting logic of fitting successive corrections to the running error [6].

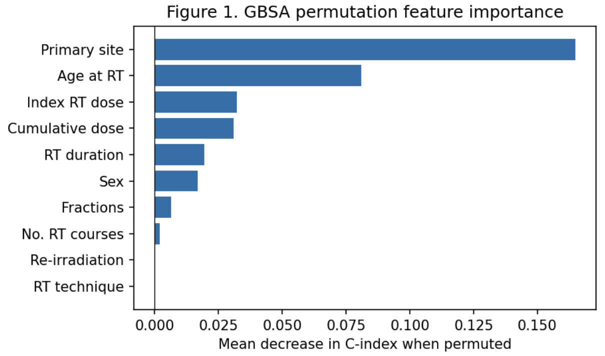
Assumptions and when it performs well. Boosting relaxes the *functional form* of the Cox risk score, allowing trees to capture nonlinearities and interactions (for example, between mandibular dose, primary site, and re-irradiation) with no parametric specification. It retains the proportional-hazards assumption: the hazard ratio between two patients is still multiplicative and constant over time. The method is well suited to data sets with substantial censoring, because the partial-likelihood loss uses censored patients rather than discarding them, and to settings where interactions among covariates are plausible but unknown. Its principal risk is overfitting when the number of events is small, which is controlled through shallow trees, a small learning rate, subsampling, and cross-validation of the number of boosting iterations. Models are evaluated with Harrell's concordance index (C-index) [7], the survival analogue of the AUC, which measures how well predicted risk ranks patients by observed event order (0.5 = chance, 1.0 = perfect).

Section 3: Architecture, Results, and Discussion

Model setup. Categorical predictors were one-hot encoded and numeric predictors passed through unchanged. We fit a gradient-boosted Cox model (`scikit-survival GradientBoostingSurvivalAnalysis` [8]) with 200 trees, learning rate $\lambda = 0.05$, maximum depth 3, and 0.8 subsampling — deliberately conservative settings given only 74 events. Because standard gradient-boosted Cox handles a single event type plus censoring, ORN was modeled as the event and death without ORN (n = 252) was treated as censoring (a cause-specific specification). Discrimination was estimated by 5-fold cross-validation stratified on event status, and benchmarked against a standard linear Cox model fit to the same features. Feature importance was computed by permutation (the mean decrease in C-index when each feature's values are shuffled), and patients were divided into predicted-risk tertiles to visualize the cumulative incidence of ORN.

Results. The gradient-boosted model achieved a cross-validated C-index of **0.570** (SD 0.114), essentially identical to the linear Cox baseline at **0.568** (SD 0.105). Permutation importance (Figure 1) identified primary tumor site as the dominant predictor by a wide margin, followed by age at RT, with index and cumulative dose and RT duration contributing modestly and the remaining features near zero. Despite the modest overall C-index, the model separated patients into clinically meaningful strata: the

36-month cumulative incidence of ORN was 0.5% in the low-risk tertile, 2.0% in the medium-risk tertile, and 6.3% in the high-risk tertile, an approximately 13-fold gradient (Figure 2).



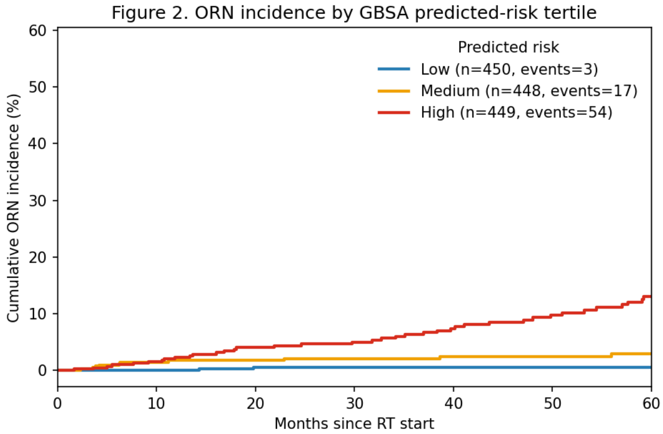


Figure 1. Permutation feature importance (mean decrease in cross-validated C-index when each predictor is shuffled). Primary tumor site and age dominate.

Figure 2. Cumulative ORN incidence by GBSA predicted-risk tertile. The tertiles separate cleanly (0.5% / 2.0% / 6.3% at 36 months) despite modest overall discrimination.

Discussion. Overall, two findings are to be discussed. First, the near-identical performance of the boosted and linear Cox models indicates that, in this cohort, the predictive signal available from baseline clinical variables is largely linear and additive; the flexibility of trees buys little when the underlying relationships are simple and the events are few. This mirrors a broader lesson that model complexity should be justified by the data, not assumed. Second, and more importantly, the modest absolute discrimination is expected rather than disappointing: the strongest mechanistic drivers of ORN (periodontal disease burden, dental health, tooth-level mandibular dose, and post-RT dental extractions) are *not* present in this baseline clinical feature set. The model therefore quantifies a clinical-only performance ceiling, and the clean risk-tertile separation shows that even coarse baseline variables (chiefly primary site and age) carry a real, if limited, risk gradient.

Several limitations are noted as follows. Death was treated as censoring, but it is properly a *competing risk*; with substantial early HNC mortality, a Fine–Gray or cause-specific competing-risks model would be the rigorous extension. The 74 events constrain model complexity and widen the cross-validated intervals. ORN is observable only while a patient remains in UCSF follow-up, so events after loss to follow-up are missed (potentially informative censoring). Finally, the study is single-institution. The clear next step is to add dental and periodontal predictors (available as structured fields in the post-2021 data, or extractable by natural language processing from earlier dental oncology notes) within a competing-risks survival framework, directly testing whether baseline dental condition improves ORN prediction beyond the clinical-only model established here.

References

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